30/06/2025 – F. Ameli, G. Lamanna, J. Pinzino, M. Sozzi

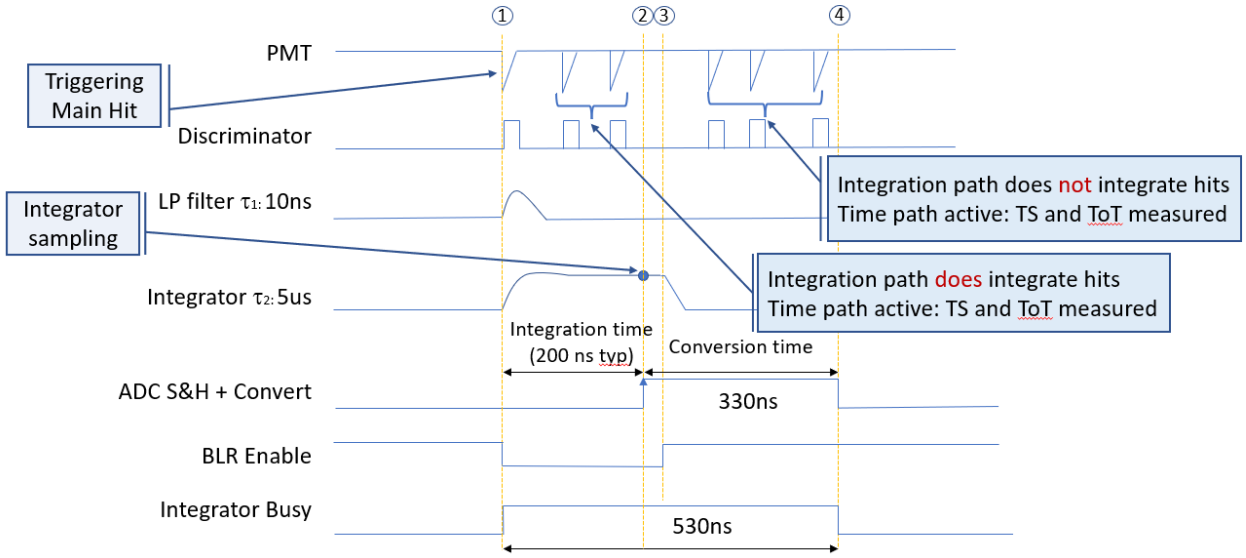
Data format from digitizer board

General assumptions

- The digitizer board contains either 12 ID channels or 10 ID + 6 OD = 16 channels. Two digitizer boards send data to a single DPB board. The data format from each digitizer contains 1 bit identifying the board. 4 bits thus are reserved for channel ID within the digitizer board.
- A single hit is uniquely assigned to a single data packet, there's no provision for packing more hits in a single data packet at the level of the digitizer. A hit is defined as having both charge (ADC) and time (TDC) information, the latter including time-over-threshold (TOT). Sub-hits are further PMT signals arriving on the same channel during the ADC digitization time or dead time, and are defined as only having TDC information; sub-hits are stored in the same data packet with the preceding hit.
- The Aurora IP core take care of the data transport to the DPB and of the data-integrity checks.
- The timing system provides, through the DPB, a common phase-coherent 125 MHz clock, a TDC-reset, and a 32-bit coarse time (both every 16.384 μs) to all the digitizers.
- The TDC-reset is used to reset the local time counter.
- The global 32-bit coarse time (16.384 μs LSB) counts the number of TDC-resets and covers a ≈20h time span, which is assumed to be bigger than a single data-taking period.
- A the common phase-coherent 125 MHz clock together with a 500 MHz clock (generated with a PLL from the 12 5MHz) generates a 45-bit local time counter (2 ns LSB, range up to ≈20h) in the digitizer. This time counter is cross checked with the global 32-bit coarse time and tdc reset arriving from the DPB.
- The low 8 bit of local time counter (with 16.384 μs LSB and range up to ≈ 4ms) are extracted and inserted in the data packet.
- The TDC works on 8-phase division of a 500 MHz clock, derived from the 125 MHz master clock.
- Hit times are measured by a combination of TDC time ("fine time") and the local coarse time value. It is generated a 16-bit fine time (start time), with 0.25 ns LSB, that covers up to 16.384 μs and a 12-bit stop time, still with 0.25 ns LSB, that covers up to 1.023 μs.
- The ADC integration time is programmable in 8.5 ns steps and his minimum values is about 200 ns: all the signals, within this time, will be integrated together.
- The ADC conversion time takes minimum 330 ns (which shadows the integrator recovery time) plus some clocks cycles of signal management; It will start just after the integration

time: All the signals arrived during the conversions would be lost, corresponding in about 350 ns of dead time

- a maximum of 15 additional sub-hits can be stored in the packet, but the sub-hits counter variable can indicate that up to 30 hits were present.



- The time-over-threshold of a single hit is proportional to the charge and it can extend up to 250 ns.
- The same time format of the main hit is used for the sub-hits: 16-bit start time, 12-bit stop time.
- Single-hit dynamic range is 1250, with required 1% resolution: charge measurement is performed by either of two 12-bit ADCs.
- The 2-bit Format field uniquely identifies the type of digitizer packet, distinguishing data packets (Format=11b), synchronization packets (Format=10b), error packets (Format=00b), and all other special packets (Format=01b) that include an 8-bit sub-format information (calibration, debugging, monitoring, etc.)
- Reserved bits are not to be used at this stage.

In the following, the different packet types are described, with some comments.

Standard data packet (format 0x3)

This is the standard format containing hits and sub-hits.

- In the reserved space of the last sub-hit a 4-bit error code can be stored, containing e.g. the information on whether some sub-hits were discarded.
- Even if the packet stores up to 15 sub-hits, the N° sub-hits variable can record a number of sub-hits up to 30.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Format Type : 2		boar d n°	ID/ OD	Channel n°:4				T-Coarse:8								T-Fine(TDC):16															
N° sub-hits:5					Err. TDC	ped	Rs	Charge:12												T-Stop(TDC):12											

Subhit-ID:4	T-Stop sub-hit 0 (TDC):12	T-Fine sub-hit 0 (TDC):16
Subhit-ID:4	T-Stop sub-hit 1 (TDC):12	T-Fine sub-hit 1 (TDC):16
Subhit-ID:4	T-Stop sub-hit 2 (TDC):12	T-Fine sub-hit 2 (TDC):16
|	|	|
|	|	|
Subhit-ID:4	T-Stop sub-hit n (TDC):12	T-Fine sub-hit n (TDC):16

FIELD NAME	Bits #	Meaning	Units	Range/Value
Format Type	2	Packet type (data, spacer, status, error, etc)	#	Value: b11
board n°	1	Board number 0 or 1 filled by the DPB	#	0-1
rg	1	Integrator range (0 = Low Gain, 1 = High Gain)	Bool	
ped	1	If the value is a pedestal	Bool	
Err. TDC	1	TDC hit missing in the data stream	Bool	
CHARGE	12	Hit Charge	#	0 – 4095
Channel n°	4	The channel number	#	0 - 11
N-SUBHIT	5	Number of sub-hits and following 32-bits word	#	0 – 30
T-COARSE	8	Low bits of Coarse Counter	16.384 us	0 – 4.194 ms
T-FINE	16	TDC time encoding hit start edge	0.250 ns	0 – 16.384 µs
T-STOP	12	Low bits of TDC time encoding hit stop edge	0.250 ns	0 – 1.023s µs
Subhit-ID	4	Cross check subhits		0x7

Pedestal data packet (format 0x3 variation of the standard one)

It is identified by the bit 25 of the second packet word.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Format Type : 2		boar d n°	ID/ OD	Channel n°:4				T-Counter:24																							
reserved					Err. pileup	Ped = 1	Reser ved	Low gain Charge:12								High gain Charge:12															

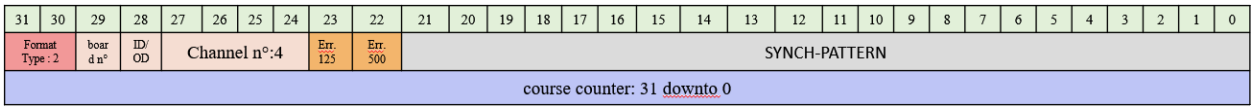
Subhit-ID:4	T-Stop sub-hit 0 (TDC):12	T-Fine sub-hit 0 (TDC):16
Subhit-ID:4	T-Stop sub-hit 1 (TDC):12	T-Fine sub-hit 1 (TDC):16
Subhit-ID:4	T-Stop sub-hit 2 (TDC):12	T-Fine sub-hit 2 (TDC):16
|	|	|
|	|	|
Subhit-ID:4	T-Stop sub-hit n (TDC):12	T-Fine sub-hit n (TDC):16

FIELD NAME	Bits #	Meaning	Units	Range/Value
Format Type	2	Packet type (data, spacer, status, error, etc)	#	Value: b11
board n°	1	Board number 0 or 1 filled by the DPB	#	0-1
Err. Pileup	1	If there is a Pileup between data and Pedestal	Bool	0-1
ped	1	If the value is a pedestal	Bool	1
Low gain Charge	12	Hit Charge	#	0 – 4095
High gain Charge	12	Hit Charge	#	0 – 4095
Channel n°	4	The channel number	#	0 - 11
N-SUBHIT	5	Number of sub-hits and following 32-bits word	#	0 – 30
T-Counter	24	Time Counter	8 ns	0 – 134 ms
T-FINE	16	TDC time encoding hit start edge	0.250 ns	0 – 16.384 µs
T-STOP	12	Low bits of TDC time encoding hit stop edge	0.250 ns	0 – 1.023s µs
Subhit-ID	4	Cross check subhits		0x7

Synchronization packet (format 0x2)

The purpose of this packet is to guarantee time synchronization, check live status and data transmission integrity, and timestamp all the following hits with coarse time information.

- 1 synchronization packet is sent every 1.048 ms (after 64 signals received from the DPB)
- 32 bits of coarse counter range from 16 µs to 20 hours
- The 8 bits of the data-packet coarse time overlap with the bits of the synchro-timestamp (from 16 µs to 4 ms)
- If the synchronization signal is not received, for example in case of a lost lock with the DPB, the synchro packet is not sent
- 2 bits of the error word are reserved for marking DPB synchronisation problems

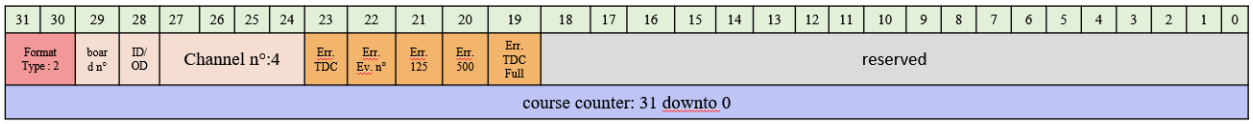


FIELD NAME	Bits #	Meaning	Units	Range/Value
Format Type	2	Packet type (data, spacer, status, error, etc)	#	Value: b10
board n°	1	Board number 0 or 1 filled by the DPB	#	0-1
Err. 500	1	Error of synchronization in the 500 MHz counter	Bool	
Err. 125	1	Error of synchronization in the 125 MHz counter	Bool	
COURSE COUNTER	32	Global course counter	16,384 µs	0 – 19,5 hours
SYNCH-PATTERN	22	22 bits used for parsing purpose (0x155555)	#	

Error packet (format 0x0)

- 5 error bits are defined
  - 2 error related to the synchronization signals (on the 500 MHz counter and on the 125MHz counter)
  - 1 error related to the missing of the TDC hit (while the ADC integration is present)
  - 1 related to the misalignment of ADC and TDC fluxes
  - 1 due to the first fifo after the TDC getting full (some tdc data are lost)

- A 32-bit course counter is included (16 µs LSB)
- The error packet is created and sent in the data stream as soon as an error is identified

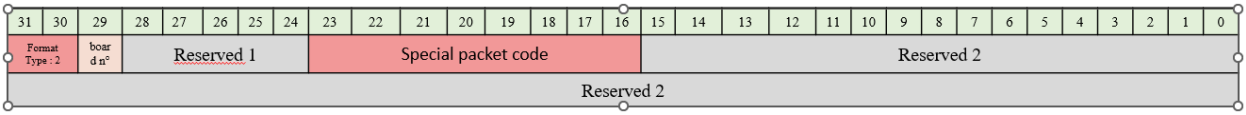


FIELD NAME	Bits #	Meaning	Units	Range/Value
Format Type	2	Packet type (data, spacer, status, error, etc)	#	Value: b00
board n°	1	Board number 0 or 1 filled by the DPB	#	0-1
Err. TDC	1	TDC hit missing in the data stream	Bool	
Err. Ev. N°	1	ADC stream and TDC stream out of sync	Bool	
Err. 125	1	Error of synchronization in the 125 MHz counter	Bool	
Err. 500	1	Error of synchronization in the 500 MHz counter	Bool	
Err. TDC Full	1	Error of lost tdc hits due to tdc fifo full	Bool	
COURSE COUNTER	32	Global course counter	16,384 µs	0 – 19,5 hours
RESERVED	20	Can be filled with others error flags	#	

Special packet (format 0x1)

This format is used to transmit all other types of information required for running and debugging, as identified by an 8-bit code called “special packet code” (256 different types of special packets, each might have a different format, including a larger size, to be defined).

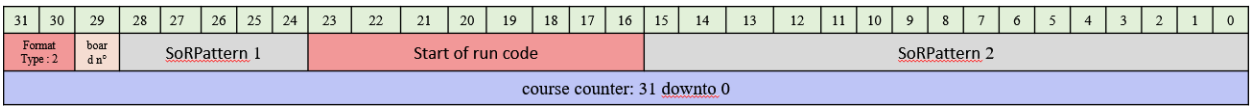
The reserved spaces can be used in different way for each kind of packet.



FIELD NAME	Bits #	Meaning	Units	Range/Value
Format Type	2	Packet type (data, spacer, status, error, etc)	#	Value: b01
board n°	1	Board number 0 or 1 filled by the DPB	#	0-1
Special packet code	8	Selector of the special packet	#	
RESERVED 1	5		#	
RESERVED 2	16		#	
RESERVED 3	32		#	

Special: start of run packet (sub-format 0x01)

This packet is generated at every start of run to guarantee a clean startup and mark the beginning of valid data.



FIELD NAME	Bits #	Meaning	Units	Range/Value
Format Type	2	Packet type (data, spacer, status, error, etc)	#	Value: b01
board n°	1	Board number 0 or 1 filled by the DPB	#	0-1
SoRPattern 1	5	Code to identify the start of run packet (no channels exist with that value)	#	0x1F
Start of run code	8	Selector of the Start of run code	#	0x01
SoRPattern 2	16	Code to identify the start of run packet	#	0x2345
COURSE COUNTER	32	Global course counter	16,382 µs	0 – 19,5 hours

Other special packets has still not implemented but we can find a possible format in the note: “Data format for the HyperK discrete digitizer” of Y. Noguchi, Y. Hayato, and F. Ameli (April 4, 2024).

In the following pages there are examples of other special packets get from SK documentations:

Class.		
BIT		Function
11	10	
0	0	QTC
0	1	TDC/DSM/SIC
1	0	Other device/block0
1	1	Other device/block1

	Class.		Device-Num				
	BIT						Function
	11	10	9	8	7	6	
QTC	0	0	0	0	0	0	QTC0 (ch0-2)
			0	0	0	1	QTC1 (ch3-5)
			0	0	1	0	QTC2 (ch6-8)
			0	0	1	1	QTC3 (ch9-11)
			0	1	0	0	QTC4 (ch12-14)
			0	1	0	1	QTC5 (ch15-17)
			0	1	1	0	QTC6 (ch18-20)
			0	1	1	1	QTC7 (ch21-23)
			1	x	x	x	not used
TDC	0	1	0	0	0	0	TDC0 (ch0-5)
					0	1	TDC1 (ch6-11)
					1	0	TDC2 (ch12-17)
					1	1	TDC3 (ch18-23)
DSM				1	0	0	DSM0 (ch0-5)
					0	1	DSM1 (ch6-11)
					1	0	DSM2 (ch12-17)
					1	1	DSM2 (ch18-23)
SIC			1	0	0	0	SIC
				0	0	1	not used
				0	1	x	not used
				1	x	x	not used
Others0	1	0	0	0	0	0	QTC-I/O control
				0	0	1	CAL control
				0	1	0	PEDESTAL control
				0	1	1	HITSUM control
				1	0	0	CLOCK control
				1	0	1	not used
				1	1	x	not used
			1	0	0	0	Thermal Surveilance
				0	0	1	Attribute Setup
				0	1	x	not used
				1	x	x	not used
Others1	1	1	0	0	x	x	not used
				1	x	x	not used
			1	0	0	0	BMC control
				0	0	1	not used
				0	1	x	not used
				1	x	x	not used

Table 1 Temperature

Temperature data is inserted when the pedestal and CAL data are taken. This gives the temperature monitor value from the sensor on the QBEE board.

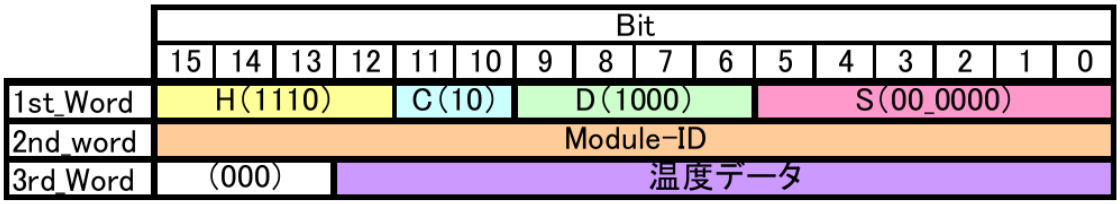


Table 2 QTC status

QTC status is inserted when the pedestal and CAL data are taken. This gives some of the setting parameters of QTC.

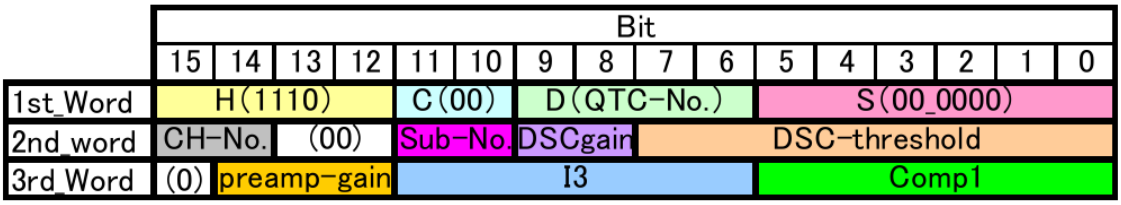


Table 3 CAL status

CAL status is inserted when the pedestal and CAL data are taken. This gives the setting of the onboard calibration source.

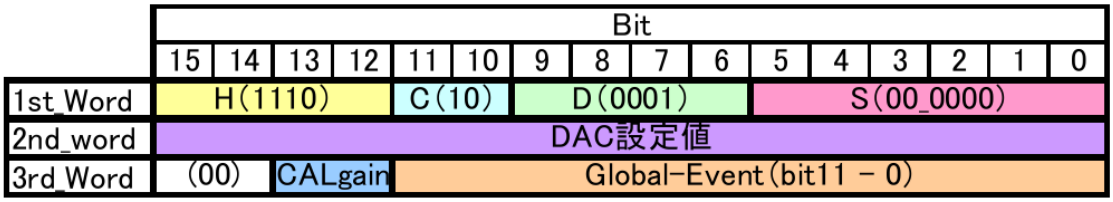


Table 4 FIFO buffer overflow

This cell is inserted when the FIFO buffer becomes full. The Global-Event is the 32bit HW counter value when the FIFO is filled up.

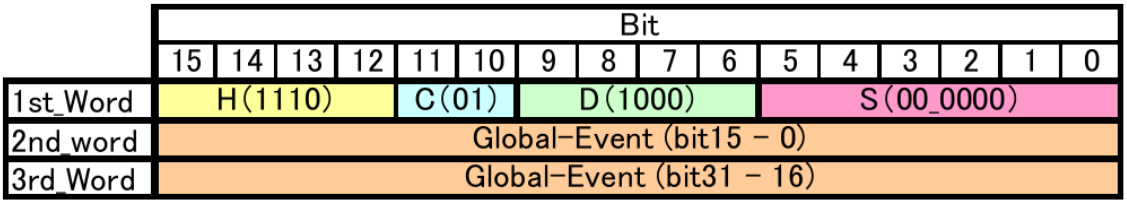


Table 5 FIFO buffer overflow recovered

This cell is inserted when the free area of FIFO buffer is more than half after the FIFO is filled up (=FIFO buffer overflow occurred).

	Bit															
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1st_Word	H(1110)				C(01)		D(1000)				S(00_0001)					
2nd_word	Global-Event (bit15 – 0)															
3rd_Word	Global-Event (bit31 – 16)															

Table 6 TDC error

If there are errors in TDC, this cell is inserted. TDC event and Error flags are provided by the TDC ASIC (AMT3).

	Bit															
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1st_Word	H(1110)				C(01)		D(TDC-No.)				S(00_0000)					
2nd_word	(0000)				TDC-Event											
3rd_Word	(00)		Error-Flag													